

Let, WLOG, $c_i \neq 0$. KIP.

$$\therefore c_1 v_1 + c_2 v_2 + \dots + c_{i-1} v_{i-1} + c_{i+1} v_{i+1} + \dots + c_p v_p = -c_i v_i$$

$$\Rightarrow v_i = \frac{c_1}{-c_i} v_1 + \frac{c_2}{-c_i} v_2 + \dots + \frac{c_{i-1}}{-c_i} v_{i-1} + \frac{c_{i+1}}{-c_i} v_{i+1} + \dots + \frac{c_p}{-c_i} v_p$$

$\therefore v_i$ is a linear combination of the other vectors of S .

Let, there exists a vector v_i from the indexed set $S = \{v_1, v_2, \dots, v_p\}$ such that v_i is a linear combination of the rest of the vectors.

$$v_i \text{ is a linear combination of the rest of the vectors.}$$

$$\therefore c_1 v_1 + c_2 v_2 + \dots + c_{i-1} v_{i-1} + c_{i+1} v_{i+1} + \dots + c_p v_p = v_i$$

$$\Rightarrow c_1 v_1 + c_2 v_2 + \dots + c_{i-1} v_{i-1} + c_{i+1} v_{i+1} + \dots + c_p v_p = 1 \cdot v_i \quad (\text{Algebraic prop viii})$$

$$\Rightarrow (c_1 v_1 + c_2 v_2 + \dots + c_{i-1} v_{i-1} + c_{i+1} v_{i+1} + \dots + c_p v_p) + (-1) \cdot v_i$$

$$= 1 \cdot v_i + (-1) \cdot v_i$$

$$\Rightarrow (c_1 v_1 + c_2 v_2 + \dots + c_{i-1} v_{i-1} + c_{i+1} v_{i+1} + \dots + c_p v_p) + (-1) \cdot v_i = (1 + (-1)) \cdot v_i$$

$$\Rightarrow (c_1 v_1 + c_2 v_2 + \dots + c_{i-1} v_{i-1} + c_{i+1} v_{i+1} + \dots + c_p v_p) + (-1) \cdot v_i = 0$$

$\therefore (-1)$ is a non-zero weight, by defn, $(v_1, v_2, \dots, v_p)$ are linearly dependent. $\square$

PRACTICE PROBLEMS

1. Let $u = \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix}$, $v = \begin{bmatrix} -6 \\ 1 \\ 7 \end{bmatrix}$, $w = \begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix}$ and $z = \begin{bmatrix} 3 \\ 7 \\ -5 \end{bmatrix}$

a) Are the sets $\{u, v\}$, $\{u, w\}$, $\{u, z\}$, $\{v, w\}$, $\{v, z\}$ and $\{w, z\}$ each linearly independent? Why or why not?

Solution:

To ensure whether a pair of vector is linearly independent, we must make sure either of the vectors are not a multiple of the other.

$\{u, v\}$: Let v be a scalar multiple of u . $\therefore \exists c \in \mathbb{R}$ s.t. $v = cu$

$$\begin{bmatrix} -6 \\ 1 \\ 7 \end{bmatrix} = c \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix} \Rightarrow \begin{bmatrix} -6 \\ 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 3c \\ 2c \\ -4c \end{bmatrix} \quad (\text{Scalar multiplication})$$

$$3c = -6 \Rightarrow c = -2$$

$$2c = 1 \Rightarrow c = \frac{1}{2}$$

This is a contradiction. $\therefore u$'s multiple is not v